

Mitch Roberts

Computer Engineer

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Relevant Experience

Co-Founder / Full Stack Web Developer

RightRenter (Startup), St. John's NL

February 2025 - Current

- Built a full-stack rental management platform from scratch using Next.js (React) and Tailwind CSS, supporting features across landlord and tenant workflows.
- Implemented location-aware listings and modular components, enabling real-time filtering.
- Integrated geolocation API to provide live user position tracking and rental discovery across various locations.

Software Developer Student

myKeen, St. John's, NL

September 2025 - December 2025

- Delivered core financial analytics features including variance dashboards, dynamic report building, AI-generated commentary, and multi-page PDF export using TypeScript, React, and SST on AWS.
- Built scalable backend services (Lambda, API Gateway, S3), integrating QuickBooks data, secure presigned uploads, and multi-company contextual workflows.
- Improved platform reliability and UX by implementing real-time AI streaming, caching layers, performance tuning, and robust authentication/authorization flows.

Software / Hardware Satellite Developer

C-CORE, St. John's NL

January 2025 - April 2025

- Developed and tested custom UART protocol in VHDL and C, supporting reliable satellite communication.
- Simulated, synthesized, and verified RTL designs using AMD Vivado, reducing debugging time.
- Contributed to ground station software for analog and digital signal processing.

May 2024 - August 2024

- Led development of a custom communications subsystem for the satellite, enabling reliable onboard data transfer.
- Created mock subsystems in Python and C to simulate hardware during integration testing and automate validation.
- Supervised and collaborated with junior co-op engineers to design and refine communication protocols.

Technical Projects

MUNStar-1 Satellite Team

Communication Subsystem Team Lead

- Designed and tested a 9-bit custom UART transmitter using VHDL for inter-module satellite communication.
- Collaborated with subsystem leads to ensure cross-module communication reliability and the satellites success.
- Led communication architecture design and mentored junior developers on protocol implementation.

TripTailor

github.com/TripTailor

Academic Project - Backend Developer (Golang)

- Developed a personalized, and dynamically updated homepage based on user selected tags for filtering
- Co-Developed a search feature that allows users to search various itineraries within our database, and select what they are looking for

Education

Bachelor of Engineering Co-Op Program - Computer

Class of 2026

Memorial University of Newfoundland - St. John's, NL

Skills

Languages:

C/C#/C++, Python, JavaScript, TypeScript, VHDL

Web:

Next.js, React, Tailwind CSS, AWS

Tools:

Git, Microsoft Suite, VS Code, Quartus, ModelSim, Vitis, Vivado, OpenAI

Soft Skills:

Leadership, Documentation, Communication, Teamwork